- REPORT

Heart Disease Analysis using Data Analytics

1. Introduction

Heart disease is one of the leading causes of death worldwide. Early prediction and analysis of heart disease can help doctors and healthcare professionals make better decisions for patient treatment.

In this project, we analyze a **Heart Disease dataset** to identify important factors such as age, cholesterol, blood pressure, and chest pain that contribute to heart disease. Data visualization and machine learning techniques are used to understand patterns in the data.

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2. Objective of the Project

The main objectives of this project are:

- To analyze patient health data related to heart disease.
- To identify key factors that influence heart disease.
- To visualize important insights using charts and dashboards.
- To build a machine learning model to predict heart disease risk.

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3. Dataset Description

The dataset used in this project contains medical information about patients.

Some important attributes in the dataset include:

Attribute	Description
Age	Age of the patient
Sex	Gender of the patient
CP	Type of chest pain
Trestbps	Resting blood pressure

Attribute	Description
Chol	Cholesterol level
FBS	Fasting blood sugar
RestECG	Resting electrocardiographic results
Thalach	Maximum heart rate achieved
Exang	Exercise induced angina
Oldpeak	ST depression
Target	Presence of heart disease

Target variable:

- **0 = No Heart Disease**
- **1 = Heart Disease**

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4. Tools and Technologies Used

The following tools were used in this project:

- **Python**
- **Pandas** – Data manipulation
- **NumPy** – Numerical operations
- **Matplotlib / Seaborn** – Data visualization
- **Scikit-learn** – Machine learning model
- **Tableau** – Dashboard creation

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5. Data Analysis and Visualization

Different visualization techniques were used to understand the dataset:

- Age distribution of patients
- Gender distribution

- Relationship between cholesterol and heart disease
- Relationship between blood pressure and heart disease
- Chest pain type analysis
- Correlation heatmap

These visualizations help in identifying patterns and trends in the data.

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6. Machine Learning Model

A **Random Forest Classifier** was used to build the prediction model.

Steps involved:

1. Data preprocessing
2. Train-test split
3. Model training
4. Model prediction
5. Model evaluation

Evaluation metrics used:

- Accuracy
- Confusion Matrix
- ROC Curve
- AUC Score

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7. Results

The machine learning model was able to predict heart disease with good accuracy.

Important factors influencing heart disease include:

- Age
- Cholesterol level
- Maximum heart rate

- Chest pain type
- Number of major vessels

These insights can help healthcare professionals in early diagnosis.

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8. Dashboard

A dashboard was created to visualize key insights such as:

- Total patients
- Heart disease cases
- Risk factors
- Age group analysis
- Gender distribution
- Cholesterol vs blood pressure

This dashboard helps in quick understanding of patient health data.

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9. Conclusion

The project successfully analyzed heart disease data and identified important risk factors affecting heart health.

Data analytics and machine learning techniques can help in early detection and prevention of heart disease, which can improve healthcare decision-making.

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10. Future Scope

Future improvements can include:

- Using larger healthcare datasets
- Applying deep learning models
- Developing a real-time prediction system
- Integrating the system with hospital management software